USN | | | | | | | | | | [22ME41]

FOURTH SEMESTER B. Tech. EXAMINATION, MAY-2024
ENGINEERING THERMODYNAMICS

Time: 3 Hours **Maximum Marks: 100**

Instructions:
i. Missing data may be suitably assumed.
ii. Use of steam tables and psychrometric chart are permitted

ANSWER ALL QUESTIONS

		PART-A			
1.	**a.**	Explain the thermodynamic systems with suitable sketches.	**L1**	CO1	**02**
	b.	Explain the Kelvin-Planck statement of thermodynamics.	**L1**	CO2	**02**
	c.	Sketch the P-T diagram for water as pure substance.	**L1**	CO3	**02**
	d.	Define the term partial pressure and give the expressions.	**L1**	CO4	**02**
	e.	Define dew point temperature & dry bulb temperature.	**L1**	CO5	**02**
		PART-B			
2.	**a.**	i. Distinguish between intensive and extensive properties.	**L2**	CO1	**05**
		ii. Define thermodynamic equilibrium. Also explain the various thermodynamic equilibriums.			
		OR			
	b.	Explain Zeroth law of thermodynamics with suitable sketch.	**L1**	CO1	**05**
3.	**a.**	Sketch P-V diagram of polytrophic process and write the expression for work transfer computation.	**L1**	CO1	**05**
		OR			
	b.	A heat engine receives heat at the rate of 1500 kJ/min and gives an output 8.2 kJ/sec. Determine: a) Thermal Efficiency and b) The rate of heat rejection.	**L3**	CO2	**05**
4.	**a.**	Discuss the reversible and irreversible processes with examples.	**L2**	CO2	**05**
		OR			
	b.	State and prove Clausius Inequality.	**L3**	CO2	**05**
5.	**a.**	Explain the terms: **i.** Dryness fraction, **ii.** wet steam and dry steam.	**L1**	CO3	**05**

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		OR			
	b.	Sketch & explain the working principle of the separating and throttling calorimeter.	**L2**	CO3	**05**
6.	**a.**	Explain compressibility factor of a real gas with suitable sketch.	**L2**	CO4	**05**
		OR			
	b.	Explain the Dalton's law of partial pressure with suitable sketch.	**L2**	CO4	**05**
7.	**a.**	The atmospheric conditions are 20°C and specific humidity of 0.0095 kg/kg of dry air. Calculate the following: a) Partial pressure of vapor and b) Relative humidity.	**L3**	CO5	**05**
		OR			
	b.	With a neat diagram explain the adiabatic heating & cooling processes.	**L2**	CO5	**05**
		PART-C			
8.	**a.**	A piston and cylinder machine containing a fluid system has a stirring device in the cylinder. The piston is frictionless and it is held down against the fluid due to the atmospheric pressure of 101.325 kPa. The stirring device is turned 10,000 revolutions with an average torque against the fluid of 1.30 N-m. Meanwhile the piston of 0.8 m diameter moves out 1 m. Find the net work-transfer for the system.	**L3**	CO1	**10**
		OR			
	b.	Discuss the steady flow energy equations for the following; **i.** Steam turbine, **ii.** Steam nozzle and **iii.** Evaporator	**L2**	CO1	**10**
9.	**a.**	A system receives 200 kJ of heat at constant volume. It rejects 70 kJ of heat at constant pressure and work transfer to the system being 50 kJ. If the system is restored to its initial state by an adiabatic process, compute the following; **i.** Work transfer during all the processes **ii.** Change in internal energy for all the processes.	**L3**	CO2	**10**
		OR			
	b.	Calculate the internal energy per kg of superheated steam at a pressure of 10 bar and a temperature of 300° C. Also find the change of internal energy, if it is expanded to 1.4 bar ad dryness fraction 0.8.	**L3**	CO3	**10**

10.	a.	Find the enthalpy of 1 kg of steam at 12 bar when steam is a) Saturated dry steam b) 22% wet and c) superheated to 250°C. Assume specific heat of the superheated steam as 2.22 kJ/kg-K	L3	CO3	10
		OR			
	b.	Derive an expression for the entropy of an ideal gas.	L2	CO3	10
11.	a.	A mixture of 0.5 kg of carbon dioxide and 0.3 kg of N_2 is compressed from P_1= 1 bar, T_1= 20°C to P_2=5 bar in a polytrophic process for which n=1.3. Calculate the following; a) Final temperature, b) work transfer, c) Heat transfer, d) Entropy change.	L3	CO3	10
		OR			
	b.	i) Discuss on Ideal gas, Real gas and Vander Waal's equation.	L2	CO4	05
		ii) Derive the Tds equations.	L2	CO4	05
12.	a.	Explain the following; **i.** Maxwell's relations and **ii.** Clausius-Clapeyron equations	L2	CO4	10
		OR			
	b.	Calculate the pressure exerted by Carbon di-oxide in a container of 1.5 cubic meter capacity when it contains 5 kg at 27°C using Ideal gas equation and Vander Waal's gas equation. Take the universal gas constant, R= 8.3144 kJ/kg-mol.K, a = 365.6 kN-m^3/kgmol^2 and b= 0.0428 m^3/kg-mol.	L3	CO4	10
13.	a.	Discuss the sensible cooling and dehumidification process with the help of Psychrometric chart.	L2	CO5	10
		OR			
	b.	The air supplied to a room of a building in winter is to be at 17°C and have a relative humidity of 60%. If the barometric pressure is 1.01325 bar, find i) Specific humidity and ii) Dew point temperature.	L3	CO5	10